

Blockchain Assignment: Customised Hyperledger Fabric network

Assignment Objective

This assignment involves setting up a Hyperledger Fabric test network using the `fabric-samples` repository. The objective is to create a Fabric network with three organizations, deploy a chaincode, and execute transactions to verify ledger consistency.

By completing this assignment, you will:

- Set up a Hyperledger Fabric environment.
- Create custom channel and deploy smart contracts (chaincode).
- Interact with the blockchain network using cli

Prerequisites

1. GitHub Account
2. Access to GitHub Codespaces
3. Basic Knowledge of Docker and Blockchain Concepts
4. Access to <https://github.com/hyperledger/fabric-samples/>

Assignment Tasks

1. Setup Hyperledger Fabric Network

- Use Hyperledger Fabric's `test-network` to create a Fabric network.
- The network should include:
 - Two peer organizations: `JIO`, `Airtel`, and one orderer organization.

- Each peer organization should have one peer node.
- The orderer organization should have one orderer node.

2. Create a Channel

- Create a channel named `consent-channel`.
- Ensure all peers from different organizations join the channel.

3. Deploy Chaincode

- Deploy the `asset-transfer` chaincode to the `consent-channel`.
- Use the chaincode from the `fabric-samples` repository.

4. Execute Transactions

- Trigger 2-3 create asset transactions using the CLI that should individual assets in the ledger of each participants/peer.

5. Query and Validate Ledger Data

- Query assets from individual peer ledgers containers.
- Verify that both peer organizations (`JIO` and `Airtel`) have identical asset details, ensuring data consistency.

Deliverables

- Github repository link with working Fabric network with the specified configurations.
- Screenshots or logs demonstrating:
 - Channel creation and peer joining.
 - Successful chaincode deployment.
 - Execution of transactions.
 - Query results showing the same asset details on both peers.

Additional Notes

- Ensure all dependencies (Docker, Fabric binaries, Node.js, Go) are installed before starting.
- The assignment should be executed within the `fabric-samples/test-network` environment.
- If any errors occur, debug using Fabric logs and command outputs.

Happy coding! 🚀